Lección 9

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April 13, 2026

Plan de trabajo

Formulación Lagrangiana

– Aplicaciones:

  * movimiento en campo central, dispersión de partículas

    ** Resolución de problemas. Problema inverso de dispersión

  * observadores no inerciales

► Nuestro Syllabus:

Nuestro Syllabus:

FIS-310:

UNIDAD 2. Formulación Lagrangiana

2.1. Variables Generalizadas

2.2. Clasificación de los Sistemas Mecánicos

2.3. Elementos de Cálculo Variacional

2.4. Ecuaciones de Lagrange

2.5. Aplicaciones: movimiento en un campo central, observadores no inerciales

FIS-345:

Unidad I: Principios Variacionales, Lagrangianos y Hamiltonianos

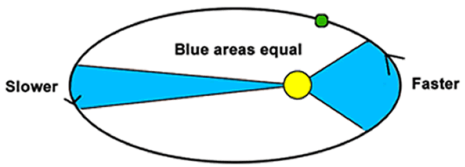
- Principio de Hamilton y Lagrangianos
- Simetrías y Leyes de Conservación (Teorema de Noether)

Unidad II: Algunas Aplicaciones

- Movimiento en campo de Fuerzas Centrales
- Movimiento de Sólido Rígido

Summary: Motion in Central field

- Motion always occurs in a plane, and the second law of Kepler is valid for any potential: a line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time



$$\int^{r} \frac{dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}\left(r\right)\right)}} = t - t_0 \quad \phi(t) = \int^{t} dt\, \frac{M}{m\, r^2(t)} \tag{1}$$

$$\phi - \phi_0 = \int^{r} dr\, \frac{M}{m\, r^2 \sqrt{\frac{2}{m}\left(E - U_{\text{eff}}\left(r\right)\right)}}$$

- Results (1-3) are *formal* solutions
  - allow to reconstruct numerically the trajectory.
  - analytical solutions known for $U(r) = \frac{\alpha}{r^n}$
    - *in terms of elementary functions for $n = 1, 2, -2$
    - **Case $n = 1$: Kepler problem: trajectories are conic sections (ellipse, parabola or hyperbola)
    - *in terms of elliptic functions for $n = -1, 3, \pm 4, \pm 6$

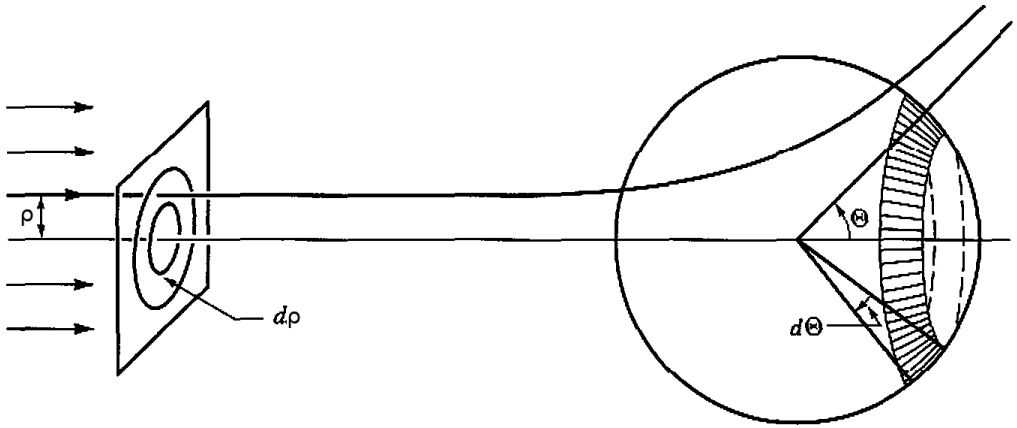
Scattering problem

Scattering in the external field

- Up to now we evaluated the shape of the orbit assuming that the initial conditions $(\boldsymbol{r}(0), \boldsymbol{v}(0))$ allow us to fix energy E, angular momentum $\boldsymbol{M}$ and the parameters of the orbit.

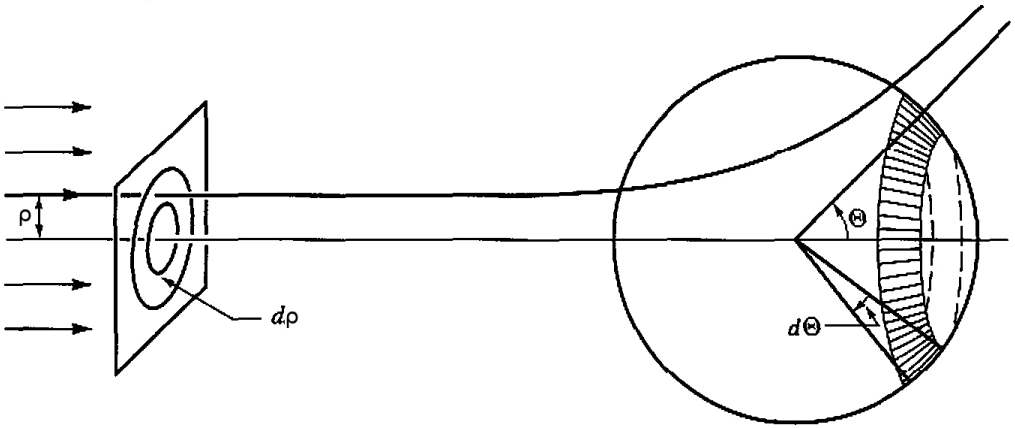
- In some problems we may now only part of this information:

−For example, we scatter a flux of particles with given energy E in the external field, and we need to find the *angular distribution* of particles after scattering. We do not know exact position $\boldsymbol{r}(0)$ to reconstruct the full trajectory of each individual particle



We expect that all particles which scatter with the same impact parameter ρ should have the same scattering angle $\Theta(\rho)$

Scattering in the external field



►

Flux of the particles J

Number of particles which pass through the unit area in transverse direction per unit of time

$$dN = J\,dS\,dt, \qquad J = \frac{dN}{dS\,dt}$$

In what follows we assume that flux J is *homogeneous*, *i.e.* does not depend on transverse coordinate

Differential cross-section

The number of particles scattered in a given solid angle $d\Omega$ or range of energies of scattered particles* $(E', E' + dE)$ and divided by the flux

$$\frac{d\sigma}{d\,\Omega} = \frac{1}{J}\frac{dN_{\text{scattered}}}{d\,\Omega} = \frac{1}{J}\frac{dN_{\text{scattered}}}{\sin\theta\,d\theta\,d\phi}$$

Scattering in the external field

- (Numerical value of) differential cross-section depends on choice of variables.

–If the two variables q and ξ are related to each other, then

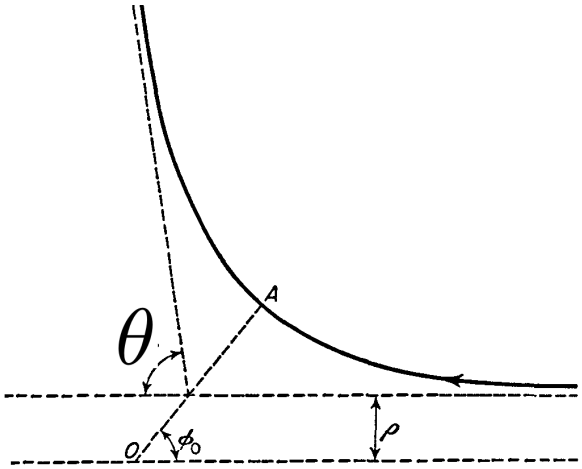
$$\frac{d\sigma}{dq} = \frac{d\sigma}{d\xi}\left|\frac{d\xi}{dq}\right|$$

–We may define differential cross-section using any generalized variables, not only angles or energies. The cross-section changes when we change variables as

$$\frac{d\sigma}{dq_1 \ldots dq_n} = \frac{d\sigma}{d\xi_1 \ldots d\xi_n}\left|\det\left(\frac{d\xi_a}{dq_b}\right)\right|$$

where $\det(\ldots)$ is a conventional Jacobian which describes change of the volume when we change coordinates. In case of just 1 variable, this is just chain rule.

Scattering in the external field

- 

  - Assume ϕ_0 is the angle between the one of the asymptotes and direction towards the closet point A, "periapsis" (in case of planets rotating around the sun it is aslo called perihelion, in case of other stars: periastron)
  - Both asymptote form the same angle ϕ_0 with direction towards
  - From geometry we may conclude that

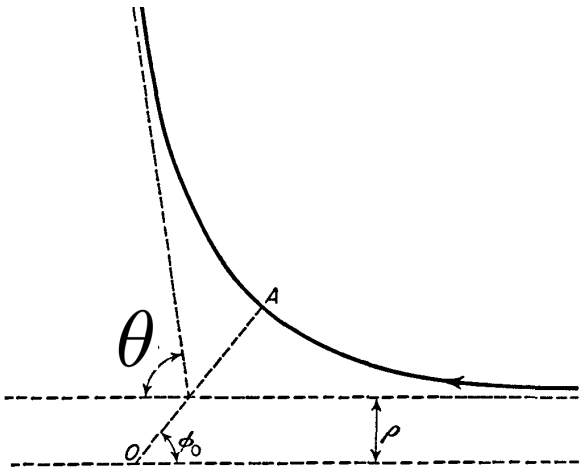
$$\theta = \pi - 2\phi_0$$

- Earlier we found the equation of trajectory in any potential in the form of indefinite integral. Adding now appropriate limits, we may get

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U(r) - \frac{M^2}{2\,m\,r^2}\right)}} = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}}, \quad U_{\text{eff}}(r) = U(r) + \frac{M^2}{2\,m\,r^2}$$

  - The value $r_{\min}$ (distance to periapsis) may be found from condition that $E = U_{\text{eff}}\left(r_{\min}\right)$, it is return point for radial motion

Scattering in the external field



$$\frac{d\sigma}{d\theta} = 2\pi\rho(\theta)\left|\frac{d\rho}{d\theta}\right|, \quad \theta = \pi - 2\phi_0$$

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}}, \quad U_{\text{eff}}(r) = U(r) + \frac{M^2}{2\,m\,r^2}$$

and $r_{\min}$ is found from

$$E = U_{\text{eff}}(r_{\min}) = U(r_{\min}) + \frac{M^2}{2\,m\,r_{\min}^2}$$

We assume that at infinity where $U(r) \approx 0$, all particles in the incident beam have the same velocity v_∞ and move in horizontal direction, so may fix the unknown parameters E, M from values at ∞:

$$E = \frac{m\,v_\infty^2}{2}, \quad M = |\boldsymbol{r} \times \boldsymbol{p}| = mv_\infty\,|\boldsymbol{r} \times \hat{\boldsymbol{x}}| = mv_\infty\rho\,|\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}}| = m\,v_\infty\,\rho,$$

$$\Rightarrow \phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{v\rho}{r^2}\,dr}{\sqrt{\frac{2}{m}\left(\frac{mv_\infty^2}{2} - U_{\text{eff}}(r)\right)}} = \int_{r_{\min}}^{\infty} \frac{\frac{v\rho}{r^2}\,dr}{\sqrt{\frac{2}{m}\left(\frac{mv_\infty^2}{2}\left(1 - \frac{\rho^2}{r^2}\right) - U(r)\right)}},$$

$$U_{\text{eff}}(r) = U(r) + \frac{mv_\infty^2}{2}\,\frac{\rho^2}{r^2}.$$

Example: Scattering in the Coulomb field

Rutherford's formula (CM frame)

In the previous lecture we analyzed in detail motion of a particle in the Coulomb field $U = -\alpha/r$. Using those results, find the cross-section of elastic scattering

$$\frac{d\sigma}{d\Omega} = \frac{1}{2\pi \sin\theta} \frac{d\sigma}{d\theta}$$

Reminder: trajectory in the Coulomb field:

$$\frac{p}{r} = 1 + e \cos\phi$$

- Case $E > 0$ (scattering): $e > 1$, trajectory is hyperbola with semiaxes

$$a = \frac{p}{e^2 - 1}, \quad b = \frac{p}{\sqrt{e^2 - 1}}$$

$$p = \frac{M^2}{m\alpha} = \frac{m v_\infty^2 \rho^2}{\alpha}, \quad e = \sqrt{1 + \frac{2EM^2}{m\alpha^2}} = \sqrt{1 + \left(\frac{m v_\infty^2 \rho}{\alpha}\right)^2}$$

Scattering in the Coulomb field

Minimal distance $r_{\min}$ which a particle may achieve:

$$E + \frac{\alpha}{r_{\min}} - \frac{M^2}{2\,m\,r_{\min}^2} = 0 \Rightarrow r_{\min} = \frac{-\alpha + \sqrt{\alpha^2 + \frac{EM^2}{2m}}}{2E} = \frac{\rho^2}{p}\,(e-1)$$

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E + \frac{\alpha}{r} - \frac{M^2}{2\,m\,r^2}\right)}} = \arccos\left(\frac{1}{e}\left(\frac{p}{r_{\min}} - 1\right)\right) - \arccos\left(-\frac{1}{e}\right) =$$

$$= \arccos\left(\frac{\alpha/mv_\infty^2\rho}{\sqrt{1 + (\alpha/mv_\infty^2\rho)^2}}\right)$$

$$\Rightarrow \rho^2 = \left(\frac{\alpha}{mv_\infty^2}\right)^2 \tan^2\phi_0 = \left(\frac{\alpha}{mv_\infty^2}\right)^2 \cot^2\frac{\theta}{2} \quad \theta = \pi - 2\phi_0,$$

$$\frac{d\sigma}{d\Omega} = \frac{d\sigma}{2\pi\sin\theta\,d\theta} = \frac{1}{2\sin\theta}\left|\frac{d\rho^2}{d\theta}\right| = \left(\frac{\alpha}{2mv_\infty^2}\right)^2 \frac{1}{\sin^4(\theta/2)}$$

(Rutherford's formula)

Scattering in the Coulomb field

$$\frac{d\sigma}{d\Omega} = \left(\frac{\alpha}{2mv_\infty^2}\right)^2 \frac{1}{\sin^4(\theta/2)}$$

- Note that

$$\sigma_{\text{tot. elast}} = \int d\Omega \, \frac{d\sigma}{d\Omega} = \infty$$

-divergence due to singular behaviour at $\theta \approx 0$ ($\rho \approx \infty$).

$\Rightarrow$this is a consequence of long-range Coulomb interaction. In any other potential which falls faster at infinity, the total scattering cross-section remains finite,

$$\sigma_{\text{tot. elast}} = \text{finite}.$$

Control question

Rutherford formula (scattering in external field):

$$\frac{d\sigma}{d\Omega} = \left(\frac{q_1 q_2}{2\mu v_\infty^2}\right)^2 \frac{1}{\sin^4(\theta/2)}$$

In all our previous considerations we assumed that flux J is homogeneous, does not depend on impact parameter ρ.

Assume that You have a beam in which the flux is not homogeneous,

$$J = J(\rho) = J_0 \exp\left(-\frac{\rho^2}{\rho_0^2}\right), \qquad J_0 = \mathrm{const}$$

but You do not know this and measured experimentally the cross-section as

$$\frac{d\sigma}{d\,\Omega_0} = \frac{1}{J_0}\frac{d\mathcal{N}_{\mathrm{scattered}}}{d\,\Omega}$$

What will be the distribution of particles which You will detect in the laboratory?

Control question (reply)

In the cross-section $d\sigma = 2\pi\rho d\rho$ the number of particles which passes through the ring per unit of time is

$$d\mathcal{N}_{\text{scattered}} = J(\rho)\, 2\pi\rho d\rho \qquad \text{instead of} \quad J_0\, 2\pi\rho d\rho$$

so in all expressions for the cross-section we should add the factor $J(\rho)/J_0$, expressed in terms of the angle θ:

$$\rho^2 = \left(\frac{\alpha}{mv_\infty^2}\right)^2 \cot^2\frac{\theta}{2}$$

so

$$J(\rho)/J_0 = \exp\left(-\left(\frac{\alpha}{mv_\infty^2\rho_0}\right)^2 \cot^2\frac{\theta}{2}\right)$$

-strong suppression at small θ

Control question

[Goldstein, 3.11]

Assume that You scatter a beams of electrons on a thin foil made of heavy ions. The charges, masses and velocities of particles in the beam and foil are given by (q_1, m_1, v_1) and $(q_2, m_2, v_2 = 0)$ respectively. The densities of particle in the beam and the foil is very low, so You may assume that each particle interacts only once with a particle of another species and leaves the beam afterwards. The velocities are not relativistic, so the interaction is dominated by the Coulomb force. Describe how to find the distribution of charged particles (in lab frame) *after* the elastic scattering.

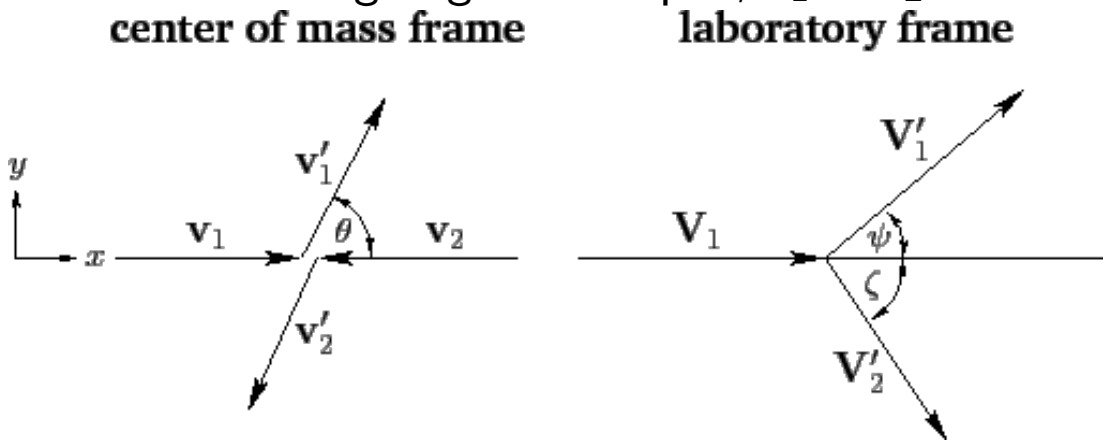
Please suggest how we can solve this problem.

Help:

– What can we say about the center of mass in this problem?

Control question (reply)

- The Coulomb force is central, internal (interaction between particles)
  $\Rightarrow$Center-of-mass will move with constant velocity in lab frame.
- In CM frame:
  – particles move always in opposite directions
  – Relative motion=motion in central field with mass $\mu = m_1 m_2 / (m_1 + m_2)$
  $\Rightarrow$their scattering angles are equal, $\theta_1 = \theta_2 = \theta$



– Scattering angle:

$$\theta_i = \arctan\left(v_y^{(i)} / v_x^{(i)}\right) \qquad (1)$$

– Use Galileo's transform to relate velocities in lab frame and CM frame

$$\boldsymbol{V}_i = \boldsymbol{v}_i + \boldsymbol{V}_{\mathrm{CM}} \qquad (2)$$

$\Rightarrow$Use (1,2) to relate the scattering angles ψ, ζ in CM frame with θ

Control question (reply)

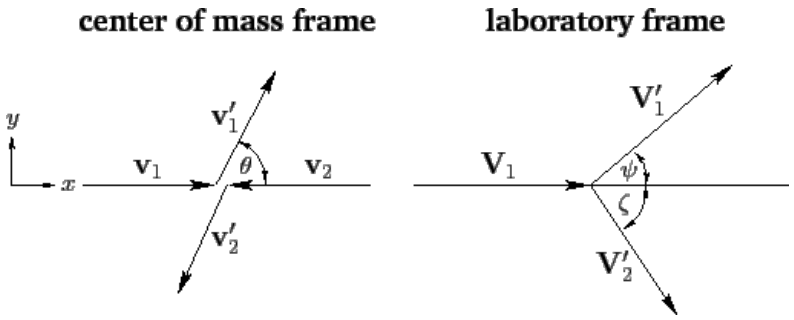
CM velocity:

$$V_{\mathrm{CM}} = \frac{m_1}{m_1 + m_2} v_1 \ \hat{\boldsymbol{x}}$$

Galileo transformation for velocities:

If $\vec{V}_{\mathrm{CM}} \equiv \dot{\vec{R}}_{\mathrm{CM}}$ is the velocity of th center of mass, and $\vec{v}_i$ are the velocities of particles in center-of-mass frame, then in the lab-frame

$$\vec{V}_i = \vec{v}_i + \vec{V}_{\mathrm{CM}}$$



$$\tan\psi = \frac{V_{1y}'}{V_{1x}'} = \frac{v_{1y}'}{v_{1x}' + V_{\mathrm{CM}}} = \frac{v_1 \sin\theta}{v_1 \cos\theta + V_{\mathrm{CM}}}$$

$$\tan\zeta = \frac{V_{2y}'}{V_{2x}'} = \frac{v_{2y}'}{v_{2x}' + V_{\mathrm{CM}}} = \frac{-|v_2| \sin\theta}{-|v_2| \cos\theta + V_{\mathrm{CM}}}$$

$$= \frac{v_1 \sin\theta}{v_1 \cos\theta - (m_2/m_1)\, V_{\mathrm{CM}}}$$

Control question (reply)

Relation between lab-frame angles (ψ, ζ) and the CM frame angle (θ),

$$\tan\psi = \frac{V'_{1y}}{V'_{1x}} = \frac{v'_{1y}}{v'_{1x} + V_{\mathrm{CM}}} = \frac{v_1 \sin\theta}{v_1 \cos\theta + V_{\mathrm{CM}}}$$

$$\tan^2\psi \left(v_1 \cos\theta + V_{\mathrm{CM}}\right)^2 = v_1^2 \left(1 - \cos^2\theta\right)$$

$$\Rightarrow \cos\theta = \cos\psi \sqrt{1 - \tan^2\psi \left(\frac{V_{\mathrm{CM}}^2 + 2 v_1 V_{\mathrm{CM}}}{v_1^2}\right)}$$

–We may relate the cross-sections in the lab-frame and CM frame :

$$d\Omega = 2\pi \sin\theta d\theta, \quad d\Omega_{\mathrm{lab}} = 2\pi \sin\psi \, d\psi$$

$$\frac{d\sigma}{d\Omega_{\mathrm{lab}}} = \frac{d\sigma}{d\Omega} \frac{d\Omega}{d\Omega_{\mathrm{lab}}} = \frac{d\sigma}{d\Omega} \frac{\sin\theta}{\sin\psi} \left|\frac{d\theta}{d\psi}\right| = \frac{d\sigma}{d\Omega} \frac{1}{\sin\psi} \left|\frac{d\cos\theta}{d\psi}\right|$$

Evaluating explicitly the derivative $d(\cos\theta)/d\psi$, we may obtain the cross-section $d\sigma/d\Omega_{\mathrm{lab}}$.

Control question 2

$$E - U\left(r_{\min}\right) - \frac{M^2}{2\,m\,r_{\min}^2} = \frac{m v_\infty^2}{2}\left(1 - \frac{\rho^2}{r_{\min}^2}\right) - U\left(r_{\min}\right) = 0 \qquad (1)$$

– Assume that solving (1), You found more than one root $r_{\min}$. What does this mean physically? Which root should You choose for scattering problem?

Control question 2 (reply)

$$E - U(r_{\min}) - \frac{M^2}{2\,m\,r_{\min}^2} = \frac{mv_\infty^2}{2}\left(1 - \frac{\rho^2}{r_{\min}^2}\right) - U(r_{\min}) = 0 \qquad (1)$$

– Assume that solving (1), You found more than one root $r_{\min}$. What does this mean physically? Which root should You choose for scattering problem?

Reply: This means that potentially we can have bound state orbit with the same energy, which correspond to motion between adjacent roots (r_i, r_{i+1}), assuming energy $E > U_{\text{eff}}$ in that region. In scattering problem particle comes from ∞, so only the largest (outermost) root is relevant.

Control question 3

$$E - U(r_{\min}) - \frac{M^2}{2\,m\,r_{\min}^2} = \frac{m v_\infty^2}{2}\left(1 - \frac{\rho^2}{r_{\min}^2}\right) - U(r_{\min}) = 0 \qquad (1)$$

- We always assumed that the root $r_{\min}$ is simple, so in the vicinity $r \approx r_{\min}$

$$E - U(r) - \frac{M^2}{2\,m\,r^2} \approx \kappa\,(r - r_{\min}) + \mathcal{O}\,(r - r_{\min})^2\,, \quad \kappa = \mathrm{const}$$

In this case we may approximate the integral near $r_{\min}$ as

$$\int_{r_{\min}}^{r} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U(r) - \frac{M^2}{2\,m\,r^2}\right)}} \approx \frac{M}{r_{\min}^2\sqrt{2\kappa m}} \int_{r_{\min}}^{r} \frac{dr}{\sqrt{r - r_{\min}}}.$$

While the *integrand* diverges, the *integral* converges/is finite.

– Assume that solving (1), for some choice of (E, M) You got double root $r_{\min}$,

$$E - U(r) - \frac{M^2}{2\,m\,r^2} \approx \lambda\,(r - r_{\min})^2 + \mathcal{O}\,(r - r_{\min})^2\,, \quad \lambda = \mathrm{const}$$

so

$$\int_{r_{\min}}^{r} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U(r) - \frac{M^2}{2\,m\,r^2}\right)}} \approx \frac{M}{r_{\min}^2\sqrt{2\lambda m}} \int_{r_{\min}}^{r} \frac{dr}{|r - r_{\min}|}.$$

has non-integrable singularity, $\phi = \infty$ (?!). What does this mean physically ???

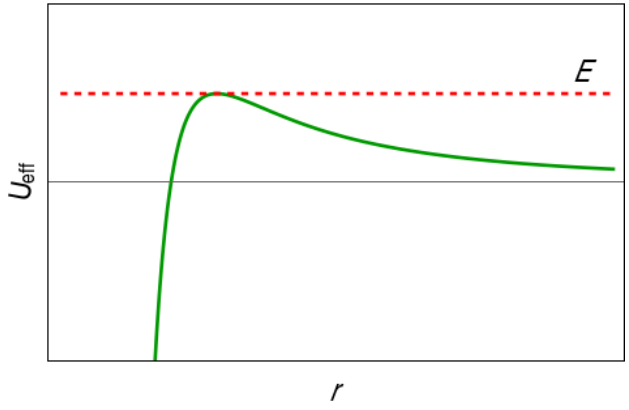
Comment 3 (reply)

This situation might happen only when energy E coincides with local extrema (maximums) of U_{eff}. In this case $\phi \to \infty$, so particle starts rotating around the center. Note that the same singularity occurs in the expression for $r(t)$,

$$t - t_0 = \int_{r_0}^{r} \frac{dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}} \sim \int_{r_{\min}}^{r} \frac{dr}{|r - r_{\min}|} = \infty$$

so the motion along such trajectory takes infinite time ($t \to \infty$).

- If the roots is of higher order, i.e. $U'_{\text{eff}}(r_{\min}) = 0$, this implies that central force (due to potential U) and centrif. force compensate each other, the particle effectively gets caught and starts rotating over circular (such exceptional orbit is called **separatrix**), see right.



- If there is no positive real-valued roots, particle falls on center

Example

Find the cross-section for scattering in the field $U = -\alpha/r^2$.

Scattering vs capture

Lecture 7: particle may fall on center in this potential under certain conditions:

– Effective potential:

$$U_{\rm eff}(r) = U(r) + \frac{M^2}{2\,m\,r^2} = \frac{\beta}{r^2}$$

$$\beta = \frac{M^2}{2m} - \alpha$$

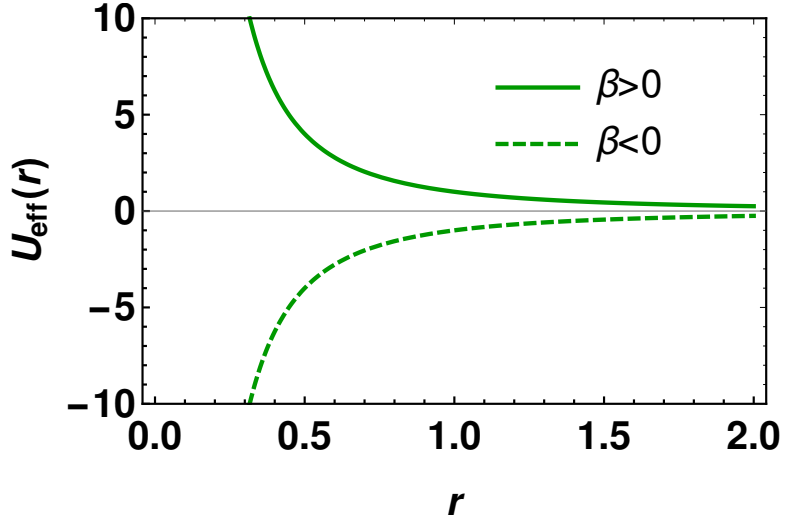
- Should distinguish cases $\beta > 0$ and $\beta < 0$:

– If $\beta > 0$, particle scatters:

  * Approaches the periapsis, and then rebounds (moves away). Distance of closest approach is the root of

$$E = U_{\rm eff}\left(r_{\rm min}\right)$$

– If $\beta < 0$, particle falls on center ($r = 0$)



Capture cross-section

- Rewrite capture condition ($\beta < 0$) in terms of ρ, v_∞:

$$\beta = \frac{M^2}{2m} - \alpha = \frac{mv_\infty^2\rho^2}{2} - \alpha < 0$$

$$\Rightarrow \rho < \rho_{\max} = \sqrt{\frac{2\alpha}{mv_\infty^2}} = \sqrt{\frac{\alpha}{E_\infty}}$$

$\Rightarrow$The field captures the particles which go close enough (with sufficiently small impact parameters). The value of "critical" (maximal) impact parameter for capture depends on energy E_∞: the higher the energy , the closer to the center it may go without getting caught.

- There is no scattering if $\rho < \rho_{\max}$, so we can only speak about total number N of caught particles
- Assume that flux J is homogeneous. Then number of particles (per unit of time) with sufficiently small impact parameters is

$$N = J \underbrace{\pi\rho_{\max}^2}_{\text{area of disk}}$$

and thus

$$\sigma_{\text{capture}} = \frac{N}{J} = \pi\rho_{\max}^2 = \frac{\pi\alpha}{E_\infty}$$

Capture cross-section - definition

$$\sigma_{\text{capture}} = \frac{N}{J}$$

where:

– The variable N is the total number of particles caught per unit of time

– The variable J is the flux (usual definition)

– If potential captures all particles with $\rho < \rho_{\max}$, then clearly

$$N = J \quad \underbrace{\pi \rho_{\max}^2}_{\text{area of disk}}$$

Scattering cross-sections in $U = -\alpha/r^2$

- For $\rho > \rho_{\max}$ particles are scattered; trajectory (Lecture 7):

$$\phi = \frac{1}{\sqrt{1 - \frac{2m\alpha}{M^2}}} \arccos\left(\frac{1}{r}\sqrt{\frac{M^2 - 2m\alpha}{2mE}}\right)$$

Can get immediately scattering angles ϕ_0, θ evaluating the difference of ϕ at $r = \infty$ and $r = r_{\min}$:

$$\phi_0 = \frac{1}{\sqrt{1 - \frac{2m\alpha}{M^2}}} \arccos\left(\frac{1}{r}\sqrt{\frac{M^2 - 2m\alpha}{2mE}}\right)\Bigg|_{r_{\min}}^{\infty}$$

The turning point ("periapsis") is found from

$$r_{\min}: \quad E - \frac{M^2 - 2m\alpha}{2m\, r_{\min}^2} = 0 \qquad r_{\min} = \sqrt{\frac{M^2 - 2m\alpha}{2m\, E}}$$

$$\Rightarrow \phi_0 = \frac{\pi}{2\sqrt{1 - \frac{\rho_{\max}^2}{\rho^2}}} = \frac{\pi - \theta}{2}$$

Scattering cross-sections in $U = -\alpha/r^2$

$$\Rightarrow \phi_0 = \frac{\pi}{2\sqrt{1 - \frac{\rho_{\max}^2}{\rho^2}}} = \frac{\pi - \theta}{2}$$

$$\Rightarrow \rho^2(\theta) = \frac{(\pi - \theta)^2 \rho_{\max}^2}{\theta(2\pi - \theta)},$$

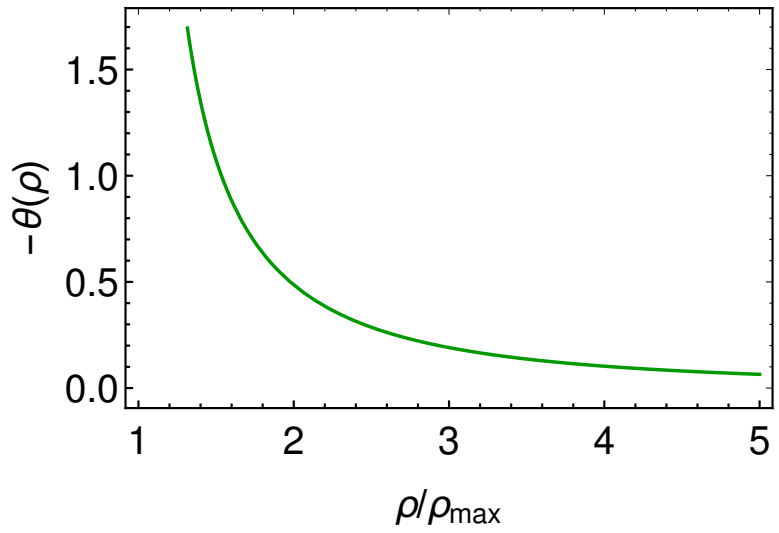
For ρ close to $\rho_{\max}$ the angle $\theta > 2\pi$
$\Rightarrow$particle "rotates" around center:

The cross-section can be found just taking derivative:

$$\Rightarrow \frac{d\sigma}{d\theta} = \pi \left| \frac{d\rho^2}{d\theta} \right| = \frac{2\pi^3 \rho_{\max}^2 \, |\pi - \theta|}{\theta^2 (2\pi - \theta)^2}$$

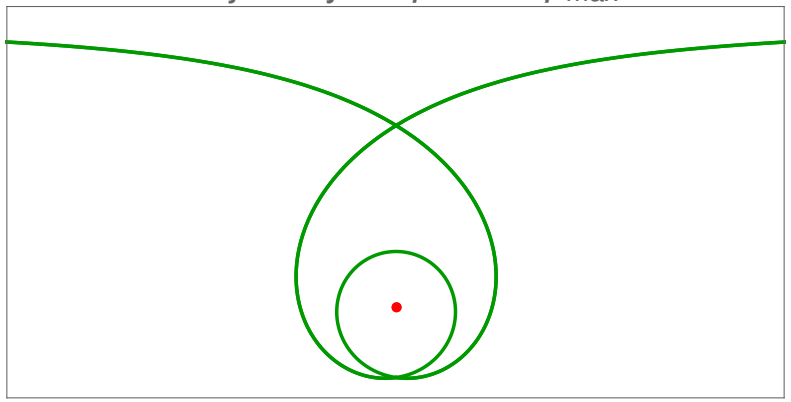
$-$peak at $\theta = 0$
$-$Vanishes at $\theta = \pi$

- Relation $\theta(\rho)$



- For $\rho \approx \rho_{\max}, \quad |\theta| > 2\pi$:

Trajectory for $\rho \approx 1.02\ \rho_{\max}$



$\Rightarrow$particle "rotates" around center:

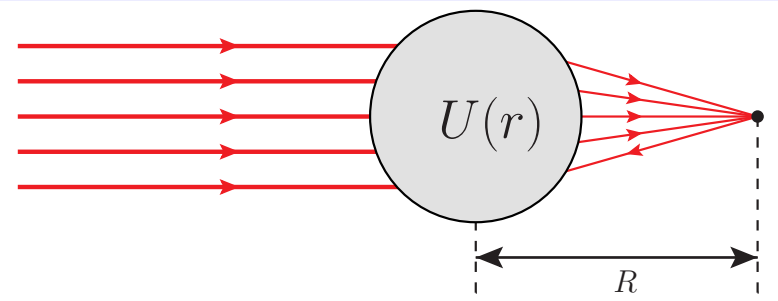
*Inverse problem of scattering (formulation)

Sometimes instead of reconstruction of trajectory or evaluation of the cross-section in known potential, we may need to do the inverse: find the unknown interaction potential between particles from experimentally measured $d\sigma/d\Omega$, or our expectations about the trajectories

(Landau, Section §18, Problem 7, *problem of Firsov*)

Assume that we know from experiment the elastic cross-section $d\sigma/d\theta$ for scattering of particles with energy E. Find the central potential $U(r)$ in which the particles scattered.

Another example of inverse problem:



A flux of monoenergetic particles with energy E scatters in the central potential $U(r)$. After the scattering all particles are focused to a point at large distance R from the center. Find the potential $U(r)$. [Luneburg lens configuration]

– Various modifications of the problem are possible (*e.g.* instead of collinear initial flux need to focus particles emitted from point source, etc)

*Inverse problem of scattering

Our standard working formulas for scattering in central potential $U(r)$:

$$\theta = \pi - 2\phi_0,$$

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U(r) - \frac{M^2}{2\,m\,r^2}\right)}} \quad (1)$$

where

$$E = \frac{m\,v_\infty^2}{2}, \quad M = m\,v_\infty\,\rho,$$

and thus

$$\frac{d\sigma}{d\theta} = 2\pi\rho(\theta)\left|\frac{d\rho}{d\theta}\right| \quad (2)$$

Scattering problem:

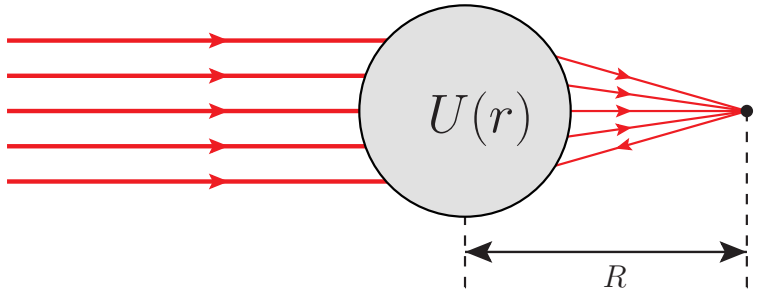
– Use (2) taking into account that $d\rho/d\theta < 0$

$$\Rightarrow \int_\theta^\pi \frac{d\sigma}{d\theta} = \pi\rho^2(\theta)$$

so if we know $d\sigma/d\theta$, we can immediately extract the dependence $\rho = \rho(\theta)$.

Focusing potentials:

– In the problem of focusing potentials we may find relation between scattering angle θ and impact parameter ρ from geometry, so we may also assume that $\theta = \theta(\rho)$ and $\rho = \rho(\theta)$ are known functions



These findings illustrate that the two problems are related to each other. So now we'll try to restore $U(r)$ from known $\rho = \rho(\theta)$.

*Inverse problem of scattering

Our standard working formulas for scattering in central potential $U(r)$:

$$\frac{\pi-\theta}{2}=\phi_0=\int_{r_{\min}}^{\infty}\frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E-U(r)-\frac{M^2}{2\,m\,r^2}\right)}}=\int_{r_{\min}}^{\infty}\frac{\frac{v_\infty\rho}{r^2}\,dr}{\sqrt{\frac{2}{m}\left(\frac{mv_\infty^2}{2}\left(1-\frac{\rho^2}{r^2}\right)-U(r)\right)}}=\quad(1)$$

$$=\int_{r_{\min}}^{\infty}\frac{\frac{1}{r^2}\,dr}{\sqrt{\left(\frac{1}{\rho^2}\left(1--\frac{2U(r)}{mv_\infty^2}\right)-\frac{1}{r^2}\right)}}$$

– Introduce

$$s=\frac{1}{r},\ x=\frac{1}{\rho^2},\ w=\sqrt{1-\frac{U}{E}}\,;$$

in new notations (1) gives

$$\Rightarrow\frac{\pi-\theta(x)}{2}=\int_0^{s_0}\frac{ds}{\sqrt{x\,w^2-s^2}}\quad(1')$$

where s_0 is the root of

$$xw^2(s_0)-s_0^2=0$$

– We'll add to rhs of (1') Heaviside step function $\Theta\left(x-s^2/w^2\right)$. This will not change result since its argument is always >0, but will help us a lot a bit later

$$\Rightarrow \frac{\pi - \theta(x)}{2} = \int_0^{s_0} \frac{ds\,\Theta\left(x - s^2/w^2\right)}{\sqrt{x\,w^2 - s^2}} \qquad (1)$$

where s_0 is the root of $$xw^2(s_0) - s_0^2 = 0$$

Now we'll multiply both parts of (1) onto $1/\sqrt{\alpha - x}$ where α is arbitrary parameter, and take integral $\int_0^\alpha dx$. In the right side of (1) change the order of integration $ds \leftrightarrow dx$ and take integral over x using *Mathematica*:

$$\int_0^\alpha \frac{\pi - \theta(x)}{2} \frac{dx}{\sqrt{\alpha - x}} = \int_0^\alpha dx \int_0^{s_0} \frac{ds\,\Theta\left(x - s^2/w^2\right)}{\sqrt{(xw^2 - s^2)(\alpha - x)}} =$$

$$= \int_0^{s_0} ds \int_{s^2/w^2}^\alpha \frac{dx}{\sqrt{(xw^2 - s^2)(\alpha - x)}} = \pi \int_0^{s_0} \frac{ds}{w}$$

Now we can rewrite the left-hand side integrating it by parts:

$$\int_0^\alpha \frac{\pi - \theta(x)}{2} \frac{dx}{\sqrt{\alpha - x}} = \frac{\pi}{2} \int_0^\alpha \frac{dx}{\sqrt{\alpha - x}} - \frac{1}{2} \int_0^\alpha \frac{\theta(x)\,dx}{\sqrt{\alpha - x}} =$$

$$= \pi\sqrt{\alpha} - \frac{1}{2} \int_0^\alpha \frac{\theta(x)\,dx}{\sqrt{\alpha - x}} = \pi\sqrt{\alpha} - \frac{1}{2} \int_0^\alpha \sqrt{\alpha - x}\,\theta'(x)\,dx$$

so we end up with

$$\pi\sqrt{\alpha} - \frac{1}{2} \int_0^\alpha \sqrt{\alpha - x}\,\theta'(x)\,dx = \pi \int_0^{s_0} \frac{ds}{w}$$

$$\pi\sqrt{\alpha} - \frac{1}{2}\int_0^{\alpha} \sqrt{\alpha - x}\theta'(x)\,dx = \pi\int_0^{s_0} \frac{ds}{w}$$

- Now we'll take d (differential) of both parts and substitute $\alpha \to s^2/w^2$, $s_0 \to s$

$$\pi d\left(s/w\right) - \frac{1}{2}d\left(s^2/w^2\right)\int_0^{s^2/w^2} \frac{\theta'(x)\,dx}{\sqrt{s^2/w^2 - x}} = \frac{\pi}{w}ds$$

$$\pi d\ln w = d\left(s/w\right)\int_0^{s^2/w^2} \frac{\theta'(x)\,dx}{\sqrt{s^2/w^2 - x}}$$

Now we'll integrate both parts and permute integrations over $d(s/w)$ and dx in right-hand side. Also we'll use

$$\int \frac{d(s/w)}{\sqrt{s^2/w^2 - x}} = \cosh^{-1}\left(\frac{\rho}{rw}\right)$$

where $\cosh^{-1}(z) \equiv \mathrm{arcosh}(z) \equiv \mathrm{acosh}(z)$

Result of integration:

(replace back $x \to 1/\rho^2$ and keep in mind that $w \equiv \sqrt{1 - \frac{U}{E}}$)

$$w = \exp\left(-\frac{1}{\pi}\int_{rw}^{\infty} \cosh^{-1}\left(\frac{\rho}{rw}\right)\frac{d\theta(\rho)}{d\rho}\,d\rho\right),$$

$$w = \exp\left(\frac{1}{\pi}\int_{rw}^{\infty} \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\,d\rho\right)$$

$\Rightarrow$algebraic equation, allows to get numerically $w(r)$ if we know $\theta(\rho)$

- In the second line we took integral by parts to replace $d\theta/d\rho$ with $\theta(\rho)$.
  - Surface term $\sim \cosh^{-1}\left(\frac{\rho}{rw}\right)\theta(\rho)$ vanishes for $\rho = rw$ due to $\cosh^{-1}(1) = 0$ and for $\rho = \infty$ since we expect $\theta(\rho = \infty) = 0$.
- Both forms are equivalent, can use any of them.

Example

Demonstrate that Rutherford's cross-section corresponds to the Coulomb's scattering potential (i.e. there is no other potentials which may give similar dependence)

Example (solution)

$$\frac{d\sigma}{d\theta} = 2\pi\,\rho(\theta)\left|\frac{d\rho}{d\theta}\right| = \left(\frac{\alpha}{2mv_\infty^2}\right)^2 \frac{2\pi\sin\theta}{\sin^4(\theta/2)}$$

$$\rho^2(\theta) = \frac{1}{2\pi}\int^\theta d\theta\,\frac{d\sigma}{d\theta} = \kappa^2\cot^2\left(\frac{\theta}{2}\right), \quad \theta(\rho) = 2\arctan\left(\frac{\kappa}{\rho}\right), \ \kappa \equiv \frac{\alpha}{mv_\infty^2}$$

$$\int_{rw}^\infty \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2w^2}}\,d\rho = \pi\sinh^{-1}\left(\frac{\kappa}{rw}\right)$$

$$w = \exp\left(\frac{1}{\pi}\int_{rw}^\infty \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2w^2}}\,d\rho\right) = \exp\left(\sinh^{-1}\left(\frac{\kappa}{rw}\right)\right)$$

Example (solution)

$$w = \exp\left(\frac{1}{\pi}\int_{rw}^{\infty}\frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\,d\rho\right) = \exp\left(\sinh^{-1}\left(\frac{\kappa}{rw}\right)\right)$$

$\Rightarrow$algebraic equation, allows to get $w(r)$:

$$\ln w = \sinh^{-1}\left(\frac{\kappa}{rw}\right),$$

$$\frac{\kappa}{rw} = \sinh(\ln w) = \frac{1}{2}\left(e^{\ln w} - e^{-\ln w}\right) = \frac{1}{2}\left(w - \frac{1}{w}\right)$$

$$\frac{2\kappa}{r} = w^2 - 1 \equiv 1 - \frac{U}{E} - 1 = -\frac{U}{E}$$

$$\Rightarrow U(r) = -\frac{2E\kappa}{r} = -\frac{\alpha}{r}, \qquad 2E\kappa = 2\frac{mv_\infty^2}{2}\frac{\alpha}{mv_\infty^2} = \alpha$$

Example 2

Let's assume ($ad\text{-}hoc$)

$$\theta = \Theta(a - \rho) \arcsin\left(\frac{\rho}{a}\right)$$

i.e. scatters (has nonzero scattering angle) only if $\rho < a$, so

$$\rho(\theta) = a \sin\theta,$$

and the cross-section

$$\frac{d\sigma}{d\theta} = 2\pi\, \rho(\theta) \left|\frac{d\rho}{d\theta}\right| = \pi a^2 \sin 2\theta$$

Find the potential $U(r)$ which corresponds to this dependence

(we'll see a bit later practical application of this result)

Solution:

$$w = \exp\left(\frac{1}{\pi}\int_{rw}^{\infty} \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\, d\rho\right)$$

$\Rightarrow$algebraic equation, allows to get $w(r)$:

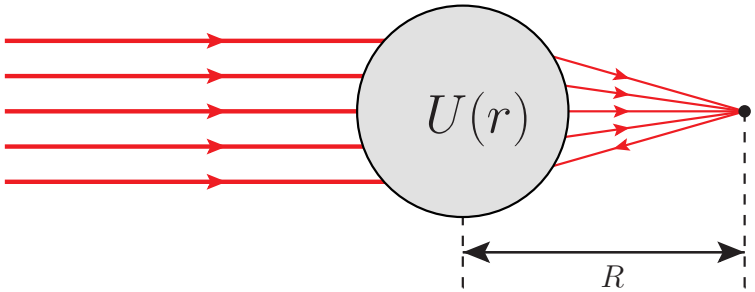
$$\frac{1}{\pi}\int_{rw}^{\infty} \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\, d\rho = -\frac{1}{2}\ln\left(\frac{a}{\sqrt{a^2 - r^2 w^2} + a}\right)$$

$$\Rightarrow w \equiv \sqrt{1 - \frac{U(r)}{E}} = \sqrt{2 - \frac{r^2}{a^2}}$$

$$U(r) = E \times \begin{cases} \left(\frac{r^2}{a^2} - 1\right), & r < a \\ 0, & r > a \end{cases}$$

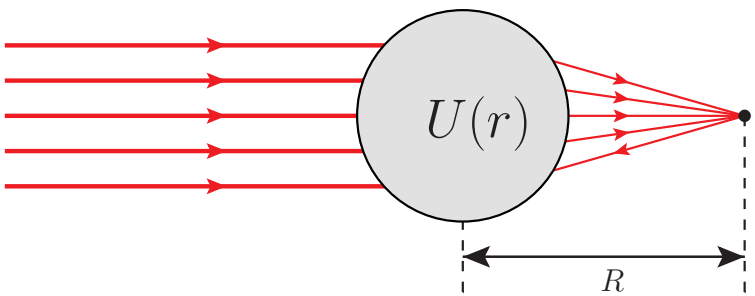
Focusing potentials

A collinear beam of monoenergetic particles with energy E and radius of the beam $\rho_{\max}$ scatters in the central potential $U(r)$. After the scattering all particles are focused to a point at large distance $R \gg \rho_{\max}$ from the center. Find the potential $U(r)$. [Luneburg lens configuration]



Different modifications/geometries are studied in the literature (*e.g.* need to focus particles from collinear beam, or emitted from point source, etc), here consider just one example

Focusing potentials



Solution:

►The impact parameter ρ is related to scattering angle θ as $\rho/R = \tan\theta$, so

$$\rho = R\,\tan\theta,\ \ \theta = \arctan\left(\frac{\rho}{R}\right)$$

⇒We can use Firsov's solution to find $U(r)$

$$w = \exp\left(\frac{1}{\pi}\int_{rw}^{\rho_{\max}} \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\, d\rho\right)$$

►Formally integral diverges if $\rho_{\max} = \infty$ ...

Comment on divergence at $\rho \to \infty$

$$\rho = R\,\tan\theta,\ \ \theta = \arctan\left(\frac{\rho}{R}\right)$$

$$w = \exp\left(\frac{1}{\pi}\int_{rw}^{\rho_{\max}} \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\,d\rho\right) \qquad (1)$$

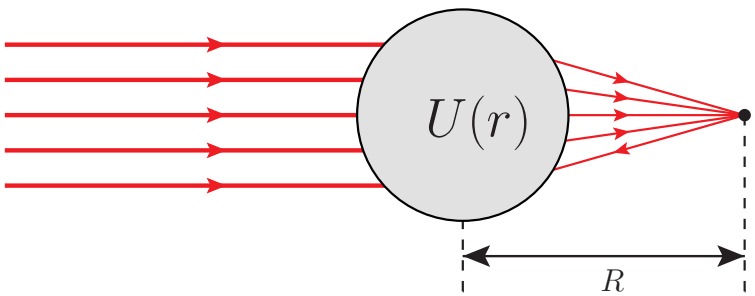
Formally integral diverges if $\rho_{\max} = \infty$. This happens because this potential requires that $\theta(\rho = \infty) \approx \pi/2$, *i.e.* even particles which go very far from center should scatter.
−Earlier in derivation of (1) we always assumed that $\theta(\rho \to \infty) = 0$.
−If we use alternative/original form of Firsov's formula

$$w = \exp\left(-\frac{1}{\pi}\int_{rw}^{\infty} \cosh^{-1}\left(\frac{\rho}{rw}\right)\frac{d\theta(\rho)}{d\rho}\,d\rho\right), \qquad (2)$$

then can see that the divergence at $\rho \to \infty$ does not happen (as You may remember, we got (1) from (2) integrating by parts and assuming that $\theta(\infty) = 0$ to suppress the surface terms)

May finalize with Mathematica, solution is elementary function (though lengthy!)

Focusing potentials



►If we consider $\rho_{\max} \ll R$ (need to focus narrow bin), then $\theta \ll 1$,

$$\tan\theta \approx \sin\theta \approx \theta$$

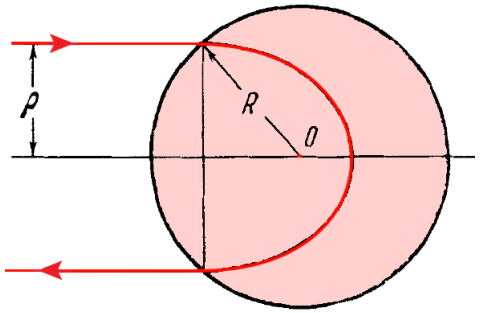
and we may use analytic result from previous problem,

$$U(r) = E \times \begin{cases} \left(\frac{r^2}{a^2} - 1\right), & r < a \\ 0, & r > a \end{cases}$$

►Our potential depends on energy (E)
–will focus (as lens) only the particles of a given energy

Example problem 3

The collinear beam of electrons of energy E_0 scatters in the central field $U(r)$. For the sake of simplicity, You may assume that the potential vanishes at distances $r \geq R$ from the center (point O). After the scattering, all the electrons with impact parameters $\rho < R$ are reflected back, as shown in the Figure.

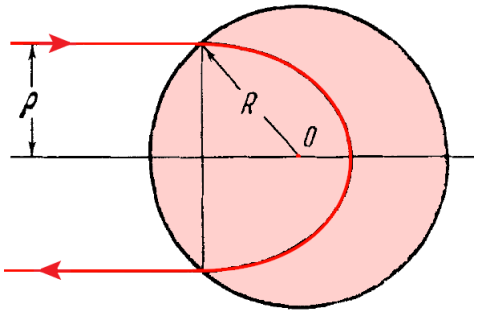


The red line with arrows is the trajectory of electrons, the colored area is the region $r < R$ where the central potential $U(r)$ is not zero.

1. Find the potential $U(r)$ which satisfies these properties.
2. A collinear beam of particles with energy $E \neq E_0$ scatters in the central potential $U(r)$ which You found earlier. Find the cross-section of elastic scattering $d\sigma/d\Omega$ for this case. Analyze what happens with the cross-section in the limit $E \to E_0$.

Goal of this problem:
Check your knowledge of inverse scattering motion in central field

Example problem 3



From geometric construction can see that the scattering angle

$$\theta(\rho < R) = \pi$$

and there is no scattering outside

This is classical inverse scattering problem,
– Since all particles have the same energy, can use Firsov's formula

$$w = \exp\left(\frac{1}{\pi}\int_{rw}^{R} \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\, d\rho\right)$$

$$\ln w = \frac{1}{\pi}\int_{rw}^{R} \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\, d\rho = \coth^{-1}\left(\frac{R}{\sqrt{R^2 - r^2 w^2}}\right),$$

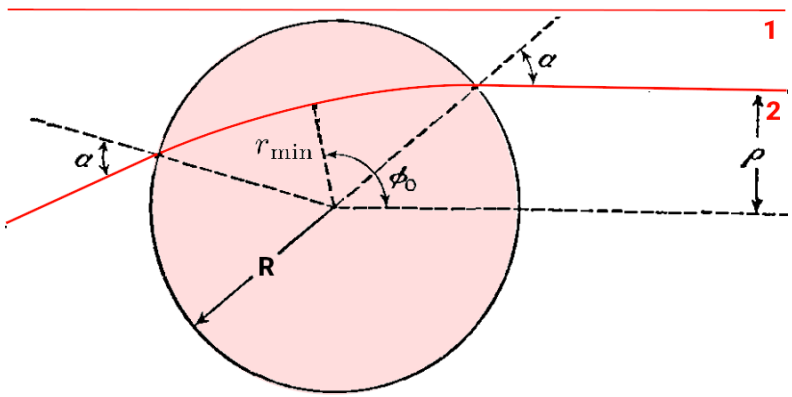
$$w = \sqrt{\frac{2R - r}{r}} = \sqrt{1 - \frac{U}{E_0}}$$

so

$$U(r < R) = 2E_0\left(1 - \frac{R}{r}\right), \quad U(r > R) = 0$$

Example problem 3

Different regimes:



Should distinguish:

- For $r > R$ trajectories are just straight lines
- For $r < R$ need to find trajectories in the potential

$$U(r) = \left(2E_0 - \frac{2E_0 R}{r}\right), \quad r < R, \quad r <$$

The first term $2E_0 = \mathrm{const}$ just "shifts" energy, does not contribute to force $\boldsymbol{F} = -\boldsymbol{\nabla} U$
The second term is ordinary Coulomb, need only to change $\alpha \to 2E_0 R$
$\Rightarrow$We may literally repeat what we discussed in Kepler's problem and get solutions as conic section

$$\frac{p}{r} = 1 + e\cos\phi, \quad p = \frac{M^2}{2mE_0 R}, \quad e = \sqrt{1 + \frac{2\left(E - 2E_0\right)M^2}{4mE_0^2 R^2}}$$

If particle's trajectory leaves the potential well, the trajectory is a mix of straight line AND conic section
– No discontinuity at the border since $U(r)$ is continuous

Example problem 3

$$\frac{p}{r} = 1 + e\cos\phi, \quad p = \frac{M^2}{2mE_0 R}, \quad e = \sqrt{1 + \frac{2\left(E - 2E_0\right)M^2}{m\alpha^2 R^2}}$$

NOTE: If You use Mathematica and try to get trajectory using general expression

$$\phi - \phi_0 = \int^r dr \, \frac{M}{m\, r^2 \sqrt{\frac{2}{m}\left(E - U_{\text{eff}}\left(r\right)\right)}}$$

You may get result in many equivalent forms:

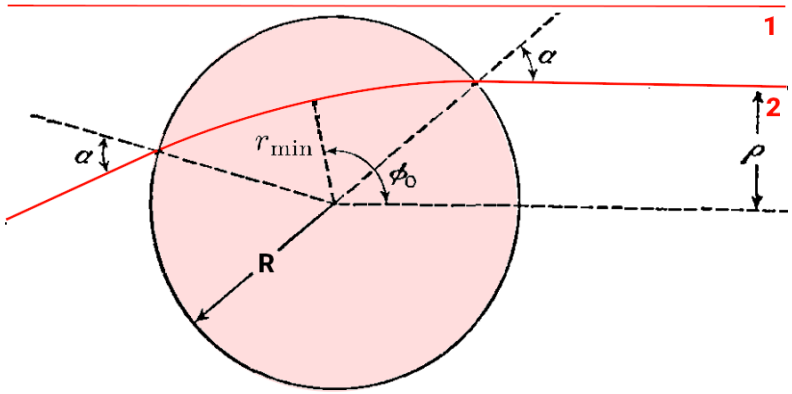
$$\phi = \arccos\left[\frac{1}{e}\left(\frac{p}{r} - 1\right)\right]$$

$$\phi = \arcsin\left[\sqrt{1 - \frac{1}{e^2}\left(\frac{p}{r} - 1\right)^2}\right]$$

$$\phi = \arctan\left[\frac{\sqrt{1 - \frac{1}{e^2}\left(\frac{p}{r} - 1\right)^2}}{\frac{1}{e}\left(\frac{p}{r} - 1\right)}\right] = \arctan\left[\frac{\sqrt{(e^2 - 1)r^2 + p^2 + 2pr}}{(p - r)}\right]$$

Example problem 3

Different regimes:



- for $\rho > R$ no scattering at all (!!!) $\Rightarrow$ $\theta = 0$, but $\phi_0 = \pi/2, \neq 0$
- For $\rho < R$ we may use conventional formulas valid in any central potential (show in blue color additional terms which differ from Kepler's problem):

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U(r) - \frac{M^2}{2\,m\,r^2}\right)}} = \int_{r_{\min}}^{\infty} \frac{\frac{\rho}{r^2}\,dr}{\sqrt{\left(1 - \frac{4E_0\Theta(R-r)}{mv_\infty^2}\left(1 - \frac{R}{r}\right) - \frac{\rho^2}{r^2}\right)}}$$

$$= \int_{r_{\min}}^{R} \frac{\frac{\rho}{r^2}\,dr}{\sqrt{\left(1 - \frac{4E_0}{mv_\infty^2}\left(1 - \frac{R}{r}\right) - \frac{\rho^2}{r^2}\right)}} + \int_{R}^{\infty} \frac{\frac{\rho}{r^2}\,dr}{\sqrt{\left(1 - \frac{\rho^2}{r^2}\right)}}$$

Minimal distance $r_{\min}$ found from equation

$$1 - \frac{4E_0\,U_0}{mv_\infty^2}\left(1 - \frac{R}{r_{\min}}\right) - \frac{\rho^2}{r_{\min}^2} = 0, \quad \rho \le R, \quad 0 \le r_{\min} \le R, \quad r_{\min} = \frac{p}{1+e}$$

Example problem 3

Evaluation of the integrals in Mathematica for $\phi_0 = (\pi - \theta)/2$:

$$\phi_0 = \sec^{-1}\left(\frac{R\sqrt{m\rho^2 v^2\left(mv^2 - 4E_0\right) + \alpha^2 R^2}}{m\rho^2 v^2 - 2E_0 R^2}\right) + \tan^{-1}\left(\frac{\rho}{\sqrt{R^2 - \rho^2}}\right)$$

To "join" several inverse functions, can apply known function (Sin, Cos, ...) use TrigExpand[...] and then FullSimplify[...], final results:

$$\cos\phi_0 = -2E_0\sqrt{\frac{R^2 - \rho^2}{m\rho^2 v_\infty^2\left(mv_\infty^2 - 4E_0\right) + 4E_0^2 R^2}}, \quad \rho = \frac{2E_0 R\sin(\phi_0)}{\sqrt{4E_0^2 + mv^2\cos^2(\phi_0)\left(mv^2 - 4E_0\right)}}$$

$$\sin\phi_0 = \frac{\rho\left(mv_\infty^2 - 2E_0\right)}{\sqrt{m\rho^2 v_\infty^2\left(mv_\infty^2 - 4E_0\right) + 4E_0^2 R^2}}, \quad \rho = \frac{2E_0 R\sin(\phi_0)}{\sqrt{\left(2E_0 - mv^2\right)^2 - mv^2\sin^2(\phi_0)\left(mv^2 - 4E\right.}}$$

$$\tan\phi_0 = \frac{\rho\left(2E_0 - mv^2\right)}{2E_0\sqrt{R^2 - \rho^2}}, \quad \rho = \frac{2E_0 R\tan(\phi_0)}{\sqrt{4E_0^2 + m^2 v^4 - 4E_0 mv^2 + 4E_0^2\tan^2(\phi_0)}}$$

All the expressions for ρ in right-hand sides are equivalent to each other
To find cross-section, can use

$$\frac{d\sigma}{d\Omega} = \frac{1}{2\pi\sin\theta}2\pi\rho\left|\frac{d\rho}{d\theta}\right| = \frac{1}{2\sin\theta}\left|\frac{d\rho^2}{d\theta}\right| = \frac{4E_0^2 R^2\left(2E_0 - mv^2\right)^2}{4\left(4E_0^2\cos^2\left(\frac{\theta}{2}\right) + \left(2E_0 - mv^2\right)^2\sin^2\left(\frac{\theta}{2}\right)\right)^2}$$

Example problem 3

$$U(r) = \begin{cases} 2E_0\left(1 - \frac{R}{r}\right), & r < R \\ 0, & r \geq R \end{cases} \qquad (1) \qquad \rho = \frac{2E_0 R\cot(\theta/2)}{\sqrt{4E_0^2 + m^2v^4 - 4E_0 mv^2 + 4E_0^2\cot^2(\theta/2)}}$$

$$\frac{d\sigma}{d\Omega} = \frac{E_0^2R^2\left(2E_0 - mv^2\right)^2}{\left(4E_0^2\cos^2\left(\frac{\theta}{2}\right) + (2E_0 - mv^2)^2\sin^2\left(\frac{\theta}{2}\right)\right)^2} = \frac{E_0^2R^2\left(E_0 - E\right)^2}{\left(E_0^2\cos^2\left(\frac{\theta}{2}\right) + (E_0 - E)^2\sin^2\left(\frac{\theta}{2}\right)\right)^2}$$

Observations and cross-check:

—If $E = mv^2/2$ approaches E_0, the cross-section vanishes (as expected)

—Our potential vanishes at large $r \to \infty$, so we don't have singularity at small scattering angles

—If we replace $E_0 \to E_0/R$ and after that consider limit $R \to \infty$, $U(r)$ turns into the Coulomb potential, and $d\sigma/d\Omega$ turns into Rutherford's formula

*Inverse problem of scattering: 2 other known solutions

Assume that we know from experiment the elastic cross-section $d\sigma/d\theta$ for particles of **fixed angular momentum** M as a function of energy E. Evaluate the potential $V(r)$.

(Problem of Hoyt)
Final result:

$$\Rightarrow \frac{\lambda}{r} = \sqrt{U} +$$
$$+ \frac{\lambda^2}{\pi\sqrt{U}} \int_{\lambda/\sqrt{U}}^{\infty} \frac{\theta(\rho)d\rho}{\rho^2\sqrt{\rho^2 - \lambda^2/U}}$$

$$\lambda = M/\sqrt{2m}, \quad U = V + \lambda^2/r^2$$

Assume that we know from experiment the elastic cross-section $d\sigma/d\theta$ for particles of **fixed impact parameter** ρ as a function of energy E. Evaluate the potential $U(r)$.

(Bogdanov-Demkov's problem)
Final result:

$$\arcsin\left(\frac{\rho}{r(U)}\right) = \frac{1}{2\pi}\int_0^U \frac{\pi - \theta(E)}{\sqrt{UE - E^2}}$$

$$U = \frac{r^2}{r^2 - \rho^2} V(r)$$

Summary

Main results:

– Concept of the differential cross-section
– Cross-section in the central field.
  * Rutherford's formula (cross-section in the Coulomb field)
  * Capture cross-section
– Inverse problem of scattering. Focusing potentials